Analysis of Daily Power Outages and Supply Reliability in India

1. Project Overview

Reliable electrical power is essential for households, industries, and public services, and frequent outages can cause operational losses and safety issues. This project analyses daily power outage data to understand the frequency, duration, and trends of interruptions. By studying variations across dates and regions, the project identifies patterns that impact supply reliability and provides insights that support better planning and improvement of power management.

Objective

- To analyse daily power outage frequency and duration.
- To identify trends and patterns in power supply reliability.
- To compare outage variations across different days or regions.
- To visualize outage data for better understanding and decision-making.

2.Data Sources

Source Description: Ministry of Power (Indiandataportal.com)

Time Period: 2017-2025

Domain: Energy

3. Problem Statement

- Frequent and unpredictable power outages affect households, industries, and public infrastructure.
- The exact frequency, duration, and patterns of outages are not clearly understood without proper data analysis.
- Lack of insights makes it difficult for authorities and industries to plan maintenance and improve grid reliability.
- There is a need to analyse daily outage data to identify trends, variations, and critical outage periods.

- Understanding these patterns can support better decision-making and help improve the stability and reliability of power supply.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Date	Date	The reporting date on which the power outage data is recorded
Region	Categorical (Text)	Geographical region where the power station is located (Northern, Southern, Eastern, etc.)
State_Name	Categorical (Text)	Name of the state affected by the power outage
State_Code	Numeric (Integer)	Unique code assigned to each state
Sector	Categorical (Text)	Ownership sector of the power station (Central, State, Private)
Power_Station_Type	Categorical (Text)	Type of power generation source (Thermal, Hydro, etc.)
Power_Station	String (Text)	Name of the power station
Power_Station_Unit	String (Text)	Specific unit of the power station where outage occurred

Outage_Type	Categorical (Text)	Nature of outage (Planned, Forced, Maintenance, etc.)
Monitored_Capacity	Numeric (Decimal)	Total power generation capacity being monitored (in MW)
Capacity_Under_Outage	Numeric (Decimal)	Power generation capacity affected due to outage (in MW)
Outage_Date	Date	Actual date on which the outage occurred
Expected_Synchronize_Date	Date	Expected date when the affected unit is planned to be synchronized back
Remark	String (Text)	Additional comments or observations related to the outage

5. Tools & Technologies

Microsoft Excel:

- Used for data cleaning, transformation, filtering, and creating Pivot Tables for preliminary analysis.

Power BI:

- Used for data modelling, DAX calculations, visualizations, and building interactive dashboards to analyze power outage trends.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning & Transformation:

- Checked and corrected data types for Date, Numeric, and Text columns.
- Removed duplicate records to avoid double counting of outage data.
- Handled missing values in critical columns such as Outage Date and Expected Synchronize Date by validating outage status.
- Standardized date formats across the dataset for consistency.
- Cleaned and standardized text fields such as Region, State Name, Power Station, and Outage Type.
- Created calculated columns such as Has_Outage_Flag to identify outage and non-outage records.

Filtering & Sorting

- Filtered out irrelevant and blank records that do not contribute to outage analysis.
- Sorted data by Date, Region, and State for better readability and analysis.
- Applied filters to focus on active outage records where capacity is under outage.

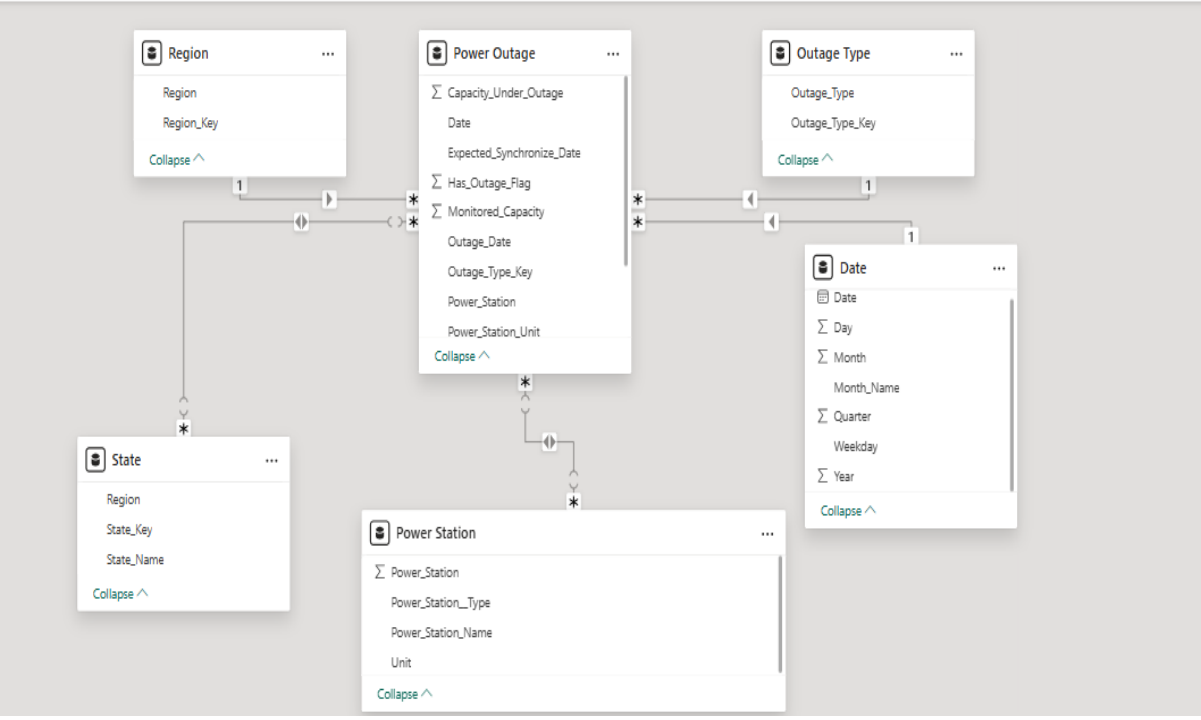
Data Model Preparation

- Converted the cleaned dataset into Fact and Dimension tables to support efficient data modelling in Power BI.
- Established clear relationships between tables to enable accurate filtering and aggregation.
- Prepared the dataset for DAX calculations and visualisation.

7. Data Modelling and DAX (Power BI)

Data Model

- Imported the cleaned dataset into Power BI Desktop.
- Created a structured data model by establishing relationships between tables.
- Defined appropriate cardinality for relationships (One-to-Many) to ensure accurate aggregation.
- Used lookup tables (such as Date, Region, State, Outage Type, and Power Station) to enable efficient filtering and slicing of data.
- Verified that all relationships flow correctly from lookup tables to the main outage data table.
- Ensured only one active date relationship is used for time-based analysis.



DAX Measures Implemented

- **Total Capacity Under Outage**

  Total cap_under outage = sum('Power Outage'[Capacity_Under_Outage])

  Calculates the total power generation capacity affected by outages.

- **Total Monitored Capacity**

  Total Monitored_capacity = sum('Power Outage'[Monitored_Capacity])

  Calculates the total monitored power generation capacity.

- **Outage Count**

  Outage_count = count('Power Outage'[Outage_Date])

  Counts the total number of outage occurrences.

- **Daily Average Outage Capacity**

  Daily Avg_Outage capacity = AVERAGE('Power Outage'[Capacity_Under_Outage])

  Calculates the average outage capacity per day.

- **No-Outage Capacity**

  No outage_capacity = [Total Monitored_capacity]-[Total cap_under outage]

  Calculates the available capacity not affected by outages.

- **Outage Percentage (%)**

  Outage_percentage(%) = DIVIDE([Total cap_under outage],[Total Monitored_capacity],0)*100

  Calculates the percentage of capacity under outage relative to monitored capacity.

- **Maximum Outage Capacity**

  Max Outage_capacity = MAX('Power Outage'[Capacity_Under_Outage])

  Identifies the highest outage capacity recorded.

- **Minimum Outage Capacity**

  Min Outage_capacity = MIN('Power Outage'[Capacity_Under_Outage])

  Identifies the lowest outage capacity recorded.

- **Yearly Outage Capacity**

  Yearly Outage_capacity = YEAR([Total cap_under outage])

  Calculates total outage capacity aggregated at the year level.

- **Year-to-Date (YTD) Capacity Under Outage.**

  YTD Capacity Under Outage = TOTALYTD([Total cap_under outage],'Date'[Date])

  Calculates cumulative outage capacity within the current year

- **Month-to-Date (MTD) Capacity Under Outage**

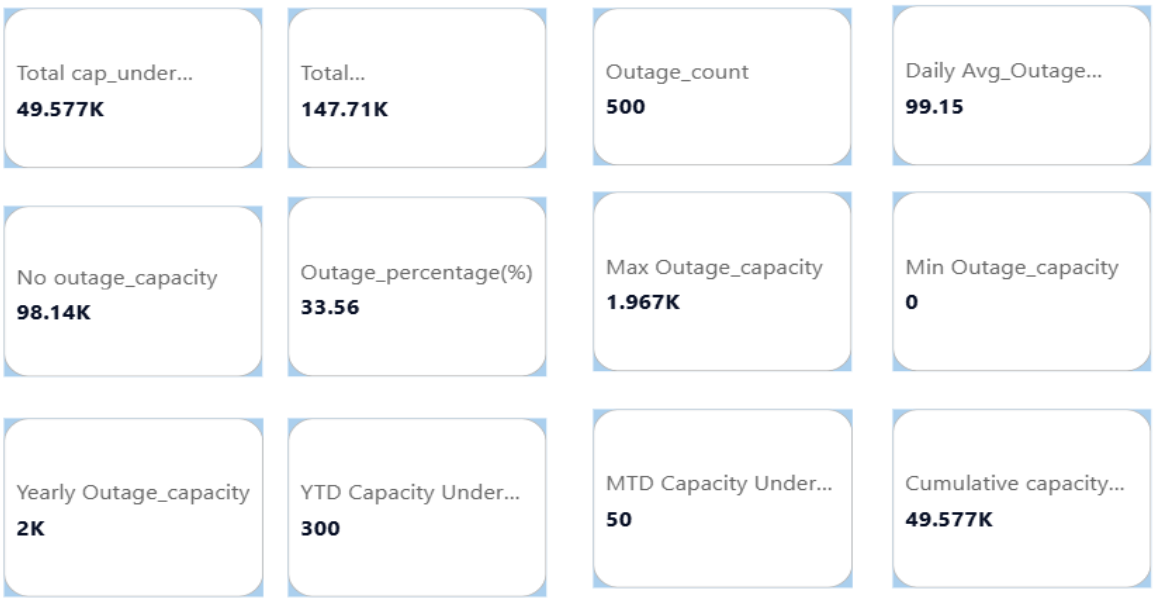
  MTD Capacity Under Outage = TOTALMTD([Total cap_under outage],'Date'[Date])

  Calculates cumulative outage capacity within the current month.

- **Cumulative Capacity Under Outage**

  Cumulative capacity under outage = CALCULATE([Totalcap_underoutage],FILTER(all('Date'[Date]),'Date'[Date]<=max('Date'[Date])))

  Calculates the running total of outage capacity over time.



8. Analysis and Visualizations (Power BI)

Dashboard Features:

Multiple Visualizations based on problem statement

Stacked Bar Chart – Region-wise Outage Capacity by Outage Type

Visual Type: Stacked Bar Chart

Purpose: Compare outage capacity across regions by outage type.

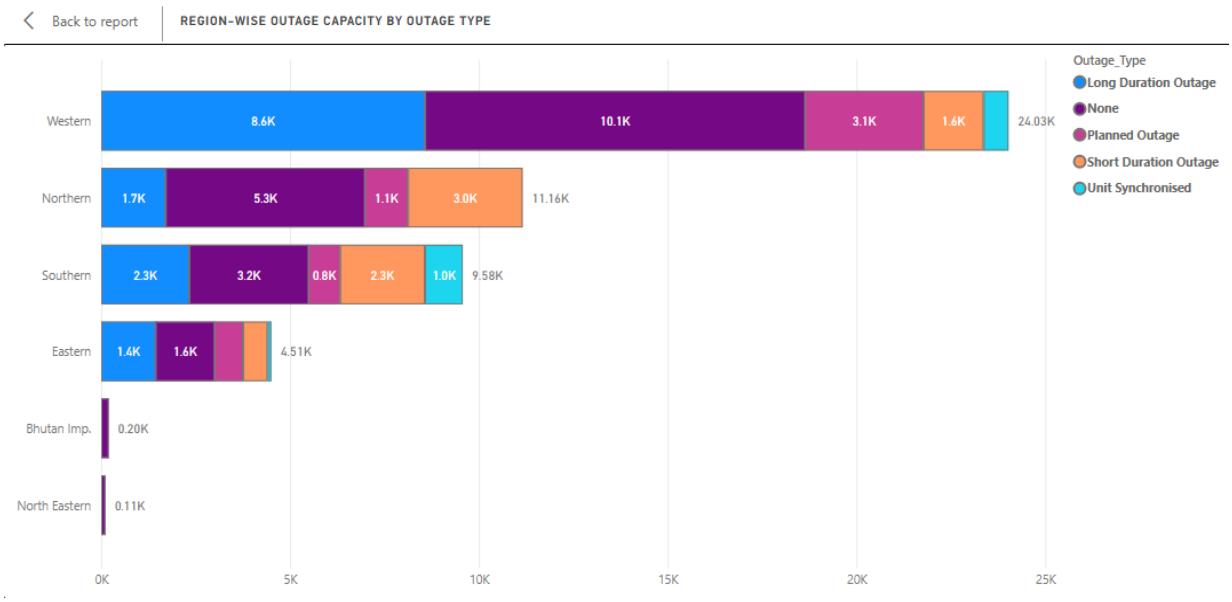
Fields Used:

Y-axis: Region

X-axis: Outage Capacity

Legend: Outage Type

Insight: The Western region has the highest outage capacity (~24.0K MW), driven mainly by Long Duration (~8.6K MW) and No-outage classified capacity (~10.1K MW). The Northern region follows with ~11.2K MW, where Short Duration outages (~3.0K MW) are relatively higher than other regions. Southern (~9.6K MW) shows a more balanced split across outage types, while Eastern (~4.5K MW) has the lowest outage among major regions; Bhutan Import (~0.2K MW) and North Eastern (~0.1K MW) contribute negligibly.



Stacked Area Chart – Monthly Trend of Outage Capacity

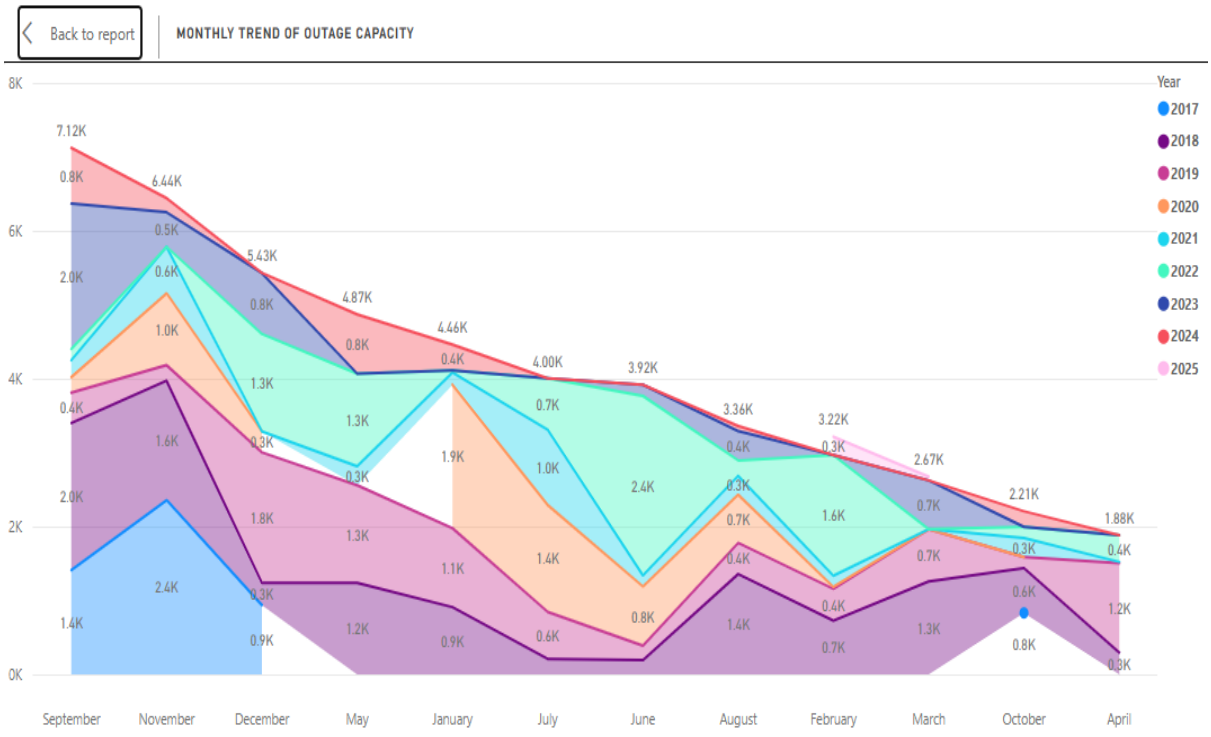
Visual Type: Stacked Area Chart

Purpose: Analyse month-wise outage capacity trend.

Fields Used:

- X-axis: Month
- Y-axis: Outage Capacity

Insight: Outage capacity peaks in September (~7.12K MW) and remains high through November–December (~6.44K–5.43K MW), indicating the worst outage period. From January onward, outages show a steady decline, reaching the lowest levels in April (~1.88K MW), clearly identifying September–December as the high-risk window and February–April as the most stable period.



Pie Chart – Outage Distribution by Outage Type

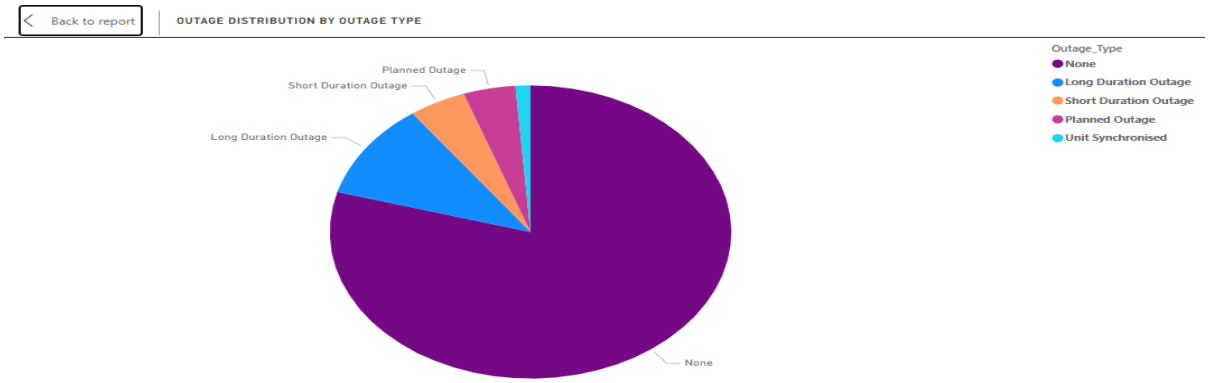
Visual Type: Pie Chart

Purpose: Display the proportional distribution of outage capacity by outage type.

Fields Used:

- Legend: Outage Type
- Values: Outage Capacity

Insight: The outage distribution is heavily skewed toward "None", which accounts for the majority (~75–80%) of total capacity, indicating most monitored capacity operates without outages. Among actual outages, Long Duration Outages (~10–12%) form the largest share, followed by Short Duration (~6–7%) and Planned Outages (~4–5%), while Unit Synchronised contributes a negligible share (<2%).



Scatter Chart – Total Monitored Capacity vs Capacity Under Outage

Visual Type: Scatter Chart

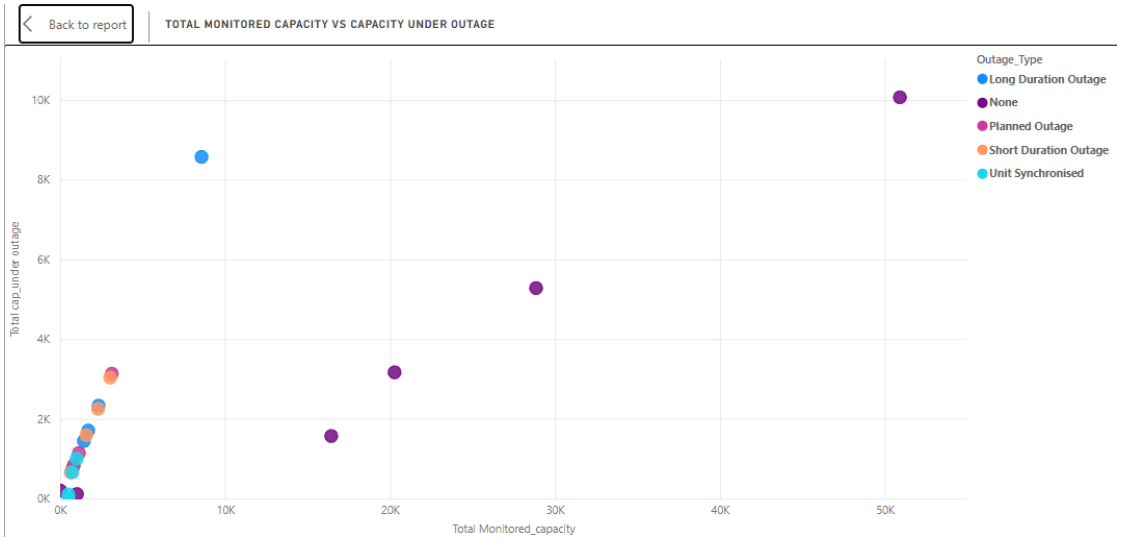
Purpose: Analyse the relationship between total monitored capacity and capacity under outage across different outage types.

Fields Used:

- X-axis: Total Monitored Capacity
- Y-axis: Total Capacity Under Outage
- Legend: Outage Type

Insight: Assets with <5K MW monitored capacity have outages mostly below 3K MW, indicating stable performance, whereas assets in the 20K–50K MW range show 3K–10K MW

outage capacity—driven largely by long-duration outages, making this range the primary contributor to overall outage impact.



Line and Clustered Column Chart – Total Cap_under Outage and Outage_Percentage (%) by Month

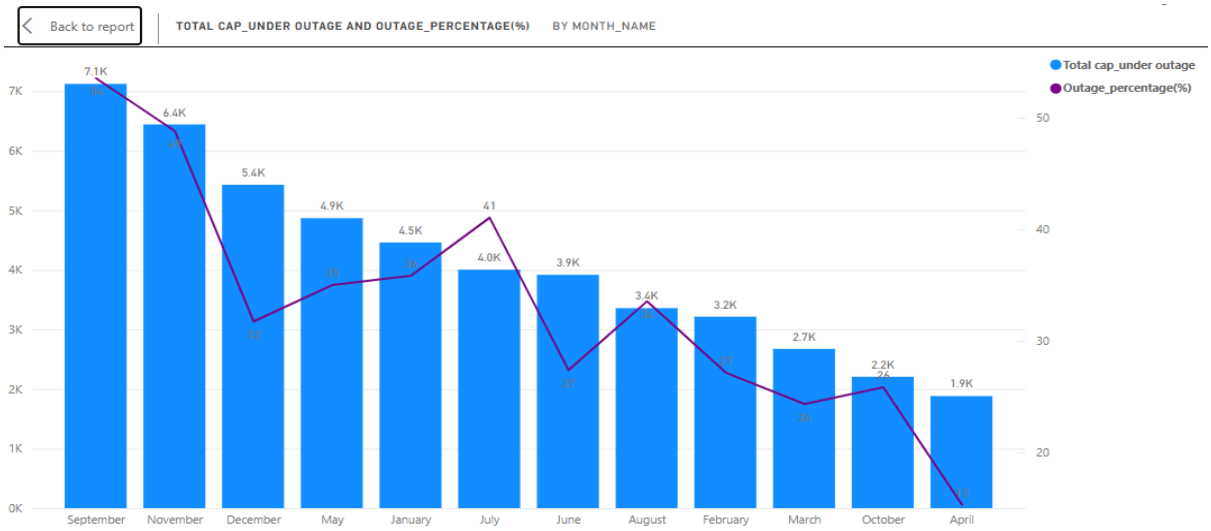
Visual Type: Line and Clustered Column Chart

Purpose: Compare total capacity under outage with outage percentage on a month-wise basis.

Fields Used:

- X-axis: Month Name
- Column Values: Total Cap_under Outage
- Line Values: Outage_Percentage (%)

Insight: September records the highest total capacity under outage (approximately 7.1K MW) and also the highest outage percentage (around 54%), clearly indicating it as the worst-performing month overall.



Map (Azure Map) – State-wise Capacity Under Outage

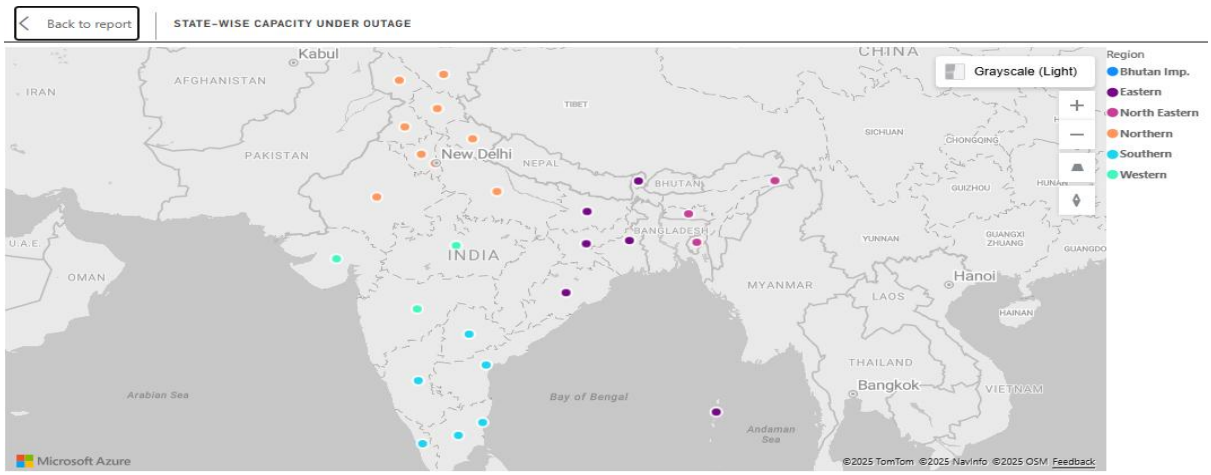
Visual Type: Map (Azure Map)

Purpose: Visualise geographical distribution of capacity under outage across different states.

Fields Used:

- Location: State_Name
- Tooltips: Total Capacity Under Outage
- Legend: Region

Insight: The map highlights region-specific outage concentration, with Northern states experiencing the highest outage impact, while Southern and Western regions show more controlled and localized outage patterns.



Gauge Chart – Outage Percentage vs SLA Target

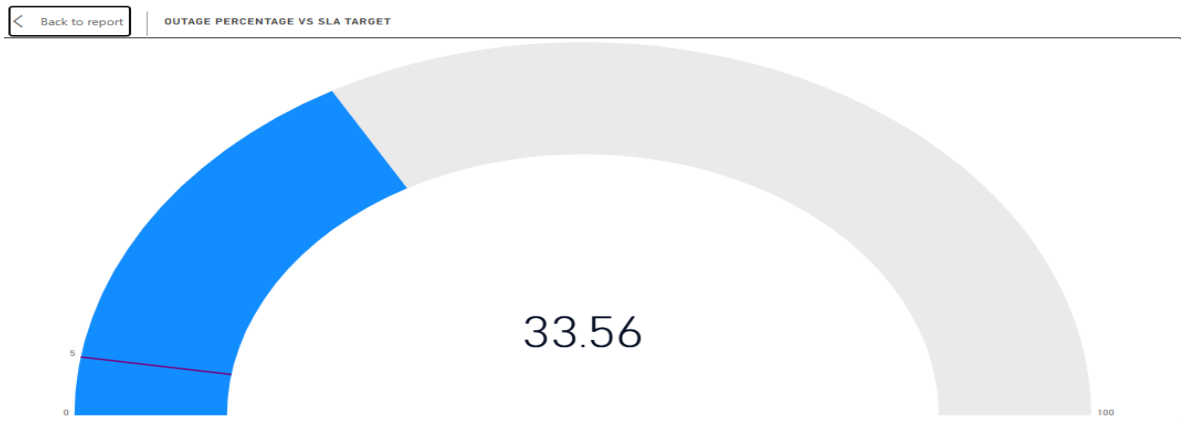
Visual Type: Gauge Chart

Purpose: Compare actual outage percentage against the defined SLA target.

Fields Used:

- Value: Outage_Percentage (%)
- Target: SLA Target (%)

Insight: The overall outage percentage of 33.56% significantly exceeds the SLA target, clearly indicating non-compliance and the need for immediate corrective actions.



KPI – Outage Percentage vs Monthly SLA Target

Visual Type: KPI

Purpose: Monitor month-wise outage percentage performance against the defined monthly SLA target.

Fields Used:

- Value: Outage_Percentage (%)
- Target: SLA Target (%)
- Trend Axis: Month_Name

Insight: The current average outage percentage is **31.74%**, which significantly exceeds the defined monthly SLA target of **5%**—almost **six times higher**—indicating continuous failure to meet SLA compliance.

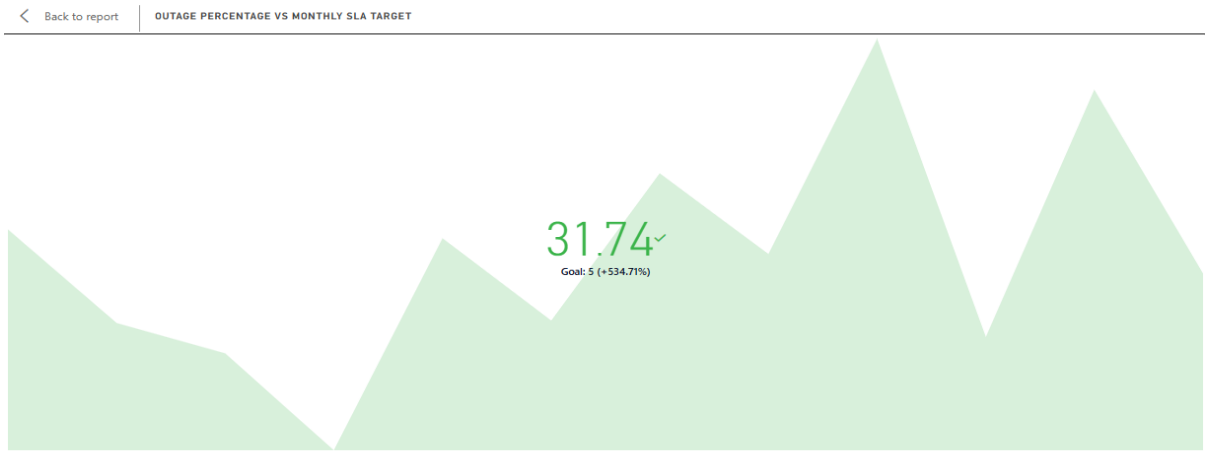


Table – State-wise Power Outage Summary

Visual Type: Table

Purpose: Provide a detailed state-wise summary of power outage performance.

Fields Used:

- Region
- State_Name
- Outage_count
- Total_cap_under_outage
- Outage_percentage (%)

Insight: A total of 49,577 MW outage across 500 records results in an overall outage percentage of 33.56%, which is well above the SLA target, indicating significant reliability issues.

Back to report | STATE-WISE POWER OUTAGE SUMMARY

Region	State_Name	Outage_count	Total cap_under outage	Outage_percentage(%)
Eastern	Andaman And Nicobar Islands	83	4505	22.78
Southern	Andhra Pradesh	109	9575	35.82
North Eastern	Arunachal Pradesh	8	110	10.76
Bhutan Imp.	Bhutan	7	198	0.00
Eastern	Bihar	83	4505	22.78
Western	Chhatisgarh	166	24028	37.00
Northern	Delhi	127	11161	31.68
Western	Gujarat	166	24028	37.00
Northern	Haryana	127	11161	31.68
Northern	Himachal Pradesh	127	11161	31.68
Northern	Jammu And Kashmir	127	11161	31.68
Eastern	Jharkhand	83	4505	22.78
Southern	Karnataka	109	9575	35.82
Southern	Kerala	109	9575	35.82
Northern	Ladakh	127	11161	31.68
Western	Madhya Pradesh	166	24028	37.00
Western	Maharashtra	166	24028	37.00
North Eastern	Meghalaya	8	110	10.76
Eastern	Odisha	83	4505	22.78
Southern	Puducherry	109	9575	35.82
Northern	Punjab	127	11161	31.68
Northern	Rajasthan	127	11161	31.68
Eastern	Sikkim	83	4505	22.78
Total		**500**	**49577**	**33.56**

Matrix – Year-wise State-level Power Outage Summary

Visual Type: Matrix

Purpose: Analyse and compare power outage performance across states on a year-wise basis.

Fields Used:

- Rows: Region, State_Name
- Columns: Year, Month
- Values:
    - Outage_count
    - Total_cap_under_outage

Insight: Outage activity increases from 2021 to 2023, peaks in 2023, then drops sharply in 2024 and becomes minimal in 2025, indicating a clear improvement in system reliability over time.

Drill-down Levels:

- Rows: Region Hierarchy + Region → State_Name
- Columns: Year → Month

Purpose: Enable hierarchical analysis of power outage data across regions, states, and time periods.

Insight: Drill-down from year to state level shows that a small set of high-capacity states drive most outages, even when overall yearly outage levels decline.

Drill on Rows

Back to report | YEAR-WISE STATE-LEVEL POWER OUTAGE SUMMARY

Year	2021	2022		2023		2024		2025		Total	
State_Name	Total cap_under outage	Outage_count	Total cap_under outage	Outage_count	Total cap_under outage	Outage_count	Total cap_under outage	Outage_count	Total cap_under outage	Outage_count	Total cap_under outage
Andaman And Nicobar Islands	849	11	210	10	570	4	600	2	50	83	4505
Andhra Pradesh	1110	8	606	13	620	17	1703	2	0	109	9575
Arunachal Pradesh	0	1	26			1	68			8	110
Bhutan	0			2	0					7	198
Bihar	849	11	210	10	570	4	600	2	50	83	4505
Chhatisgarh	210	23	6471	17	2418	10	0	3	250	166	24028
Delhi	961	19	907	9	859	6	0	1	0	127	11161
Gujarat	210	23	6471	17	2418	10	0	3	250	166	24028
Haryana	961	19	907	9	859	6	0	1	0	127	11161
Himachal Pradesh	961	19	907	9	859	6	0	1	0	127	11161
Jammu And Kashmir	961	19	907	9	859	6	0	1	0	127	11161
Jharkhand	849	11	210	10	570	4	600	2	50	83	4505
Karnataka	1110	8	606	13	620	17	1703	2	0	109	9575
Kerala	1110	8	606	13	620	17	1703	2	0	109	9575
Ladakh	961	19	907	9	859	6	0	1	0	127	11161
Madhya Pradesh	210	23	6471	17	2418	10	0	3	250	166	24028
Maharashtra	210	23	6471	17	2418	10	0	3	250	166	24028
Meghalaya	0	1	26			1	68			8	110
Odisha	849	11	210	10	570	4	600	2	50	83	4505
Puducherry	1110	8	606	13	620	17	1703	2	0	109	9575
Punjab	961	19	907	9	859	6	0	1	0	127	11161
Rajasthan	961	19	907	9	859	6	0	1	0	127	11161
Sikkim	849	11	210	10	570	4	600	2	50	83	4505
Tamil Nadu	1110	8	606	13	620	17	1703	2	0	109	9575
Telangana	1110	8	606	13	620	17	1703	2	0	109	9575
Tripura	0	1	26			1	68			8	110
Uttar Pradesh	961	19	907	9	859	6	0	1	0	127	11161
Uttarakhand	961	19	907	9	859	6	0	1	0	127	11161
West Bengal	849	11	210	10	570	4	600	2	50	83	4505
Total	**3130**	**62**	**8220**	**51**	**4467**	**38**	**2371**	**8**	**300**	**500**	**49577**

Treemap – Total Capacity Under Outage by Region and State Name

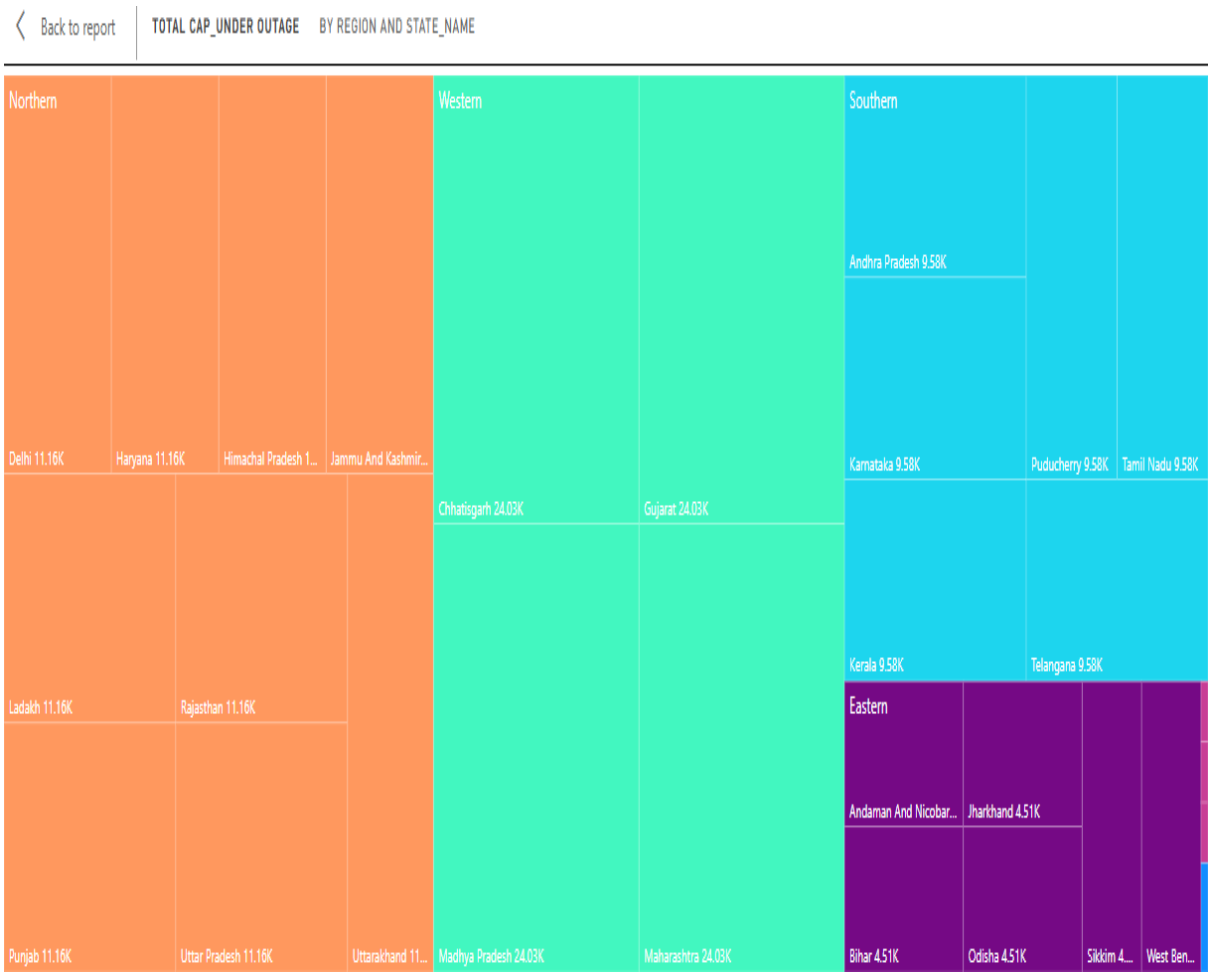
Visual Type: Treemap

Purpose: Visualize the distribution of total capacity under outage across regions and states.

Fields Used:

- Category: Region
- Details: State_Name
- Values: Total_cap_under_outage

Insight: The Western region contributes the highest total capacity under outage (~24,028 MW), making it the largest driver of overall outage impact among all regions.



Waterfall Chart – Total Capacity Under Outage by Month Name and Outage Type

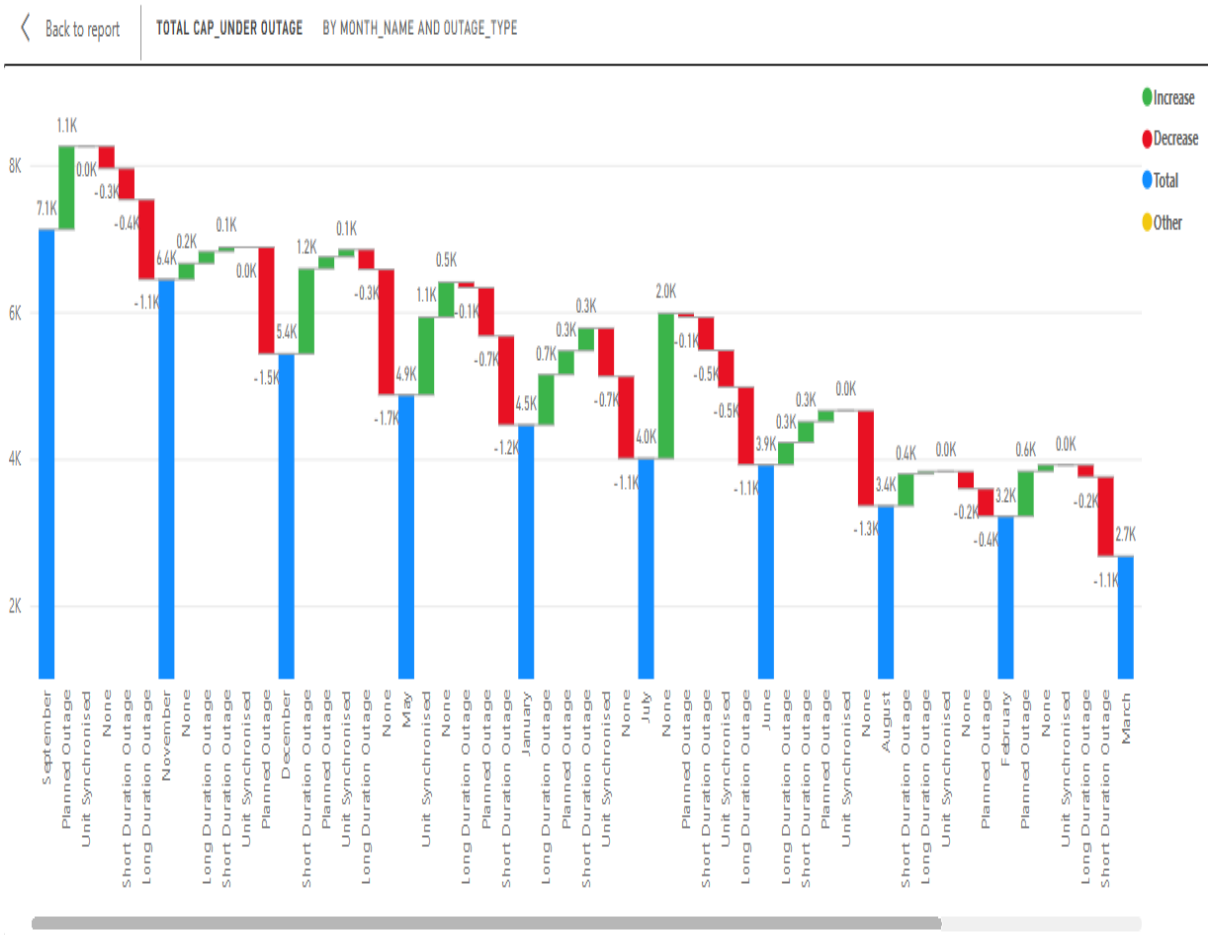
Visual Type: Waterfall Chart

Purpose: Analyse month-wise changes in total capacity under outage by outage type.

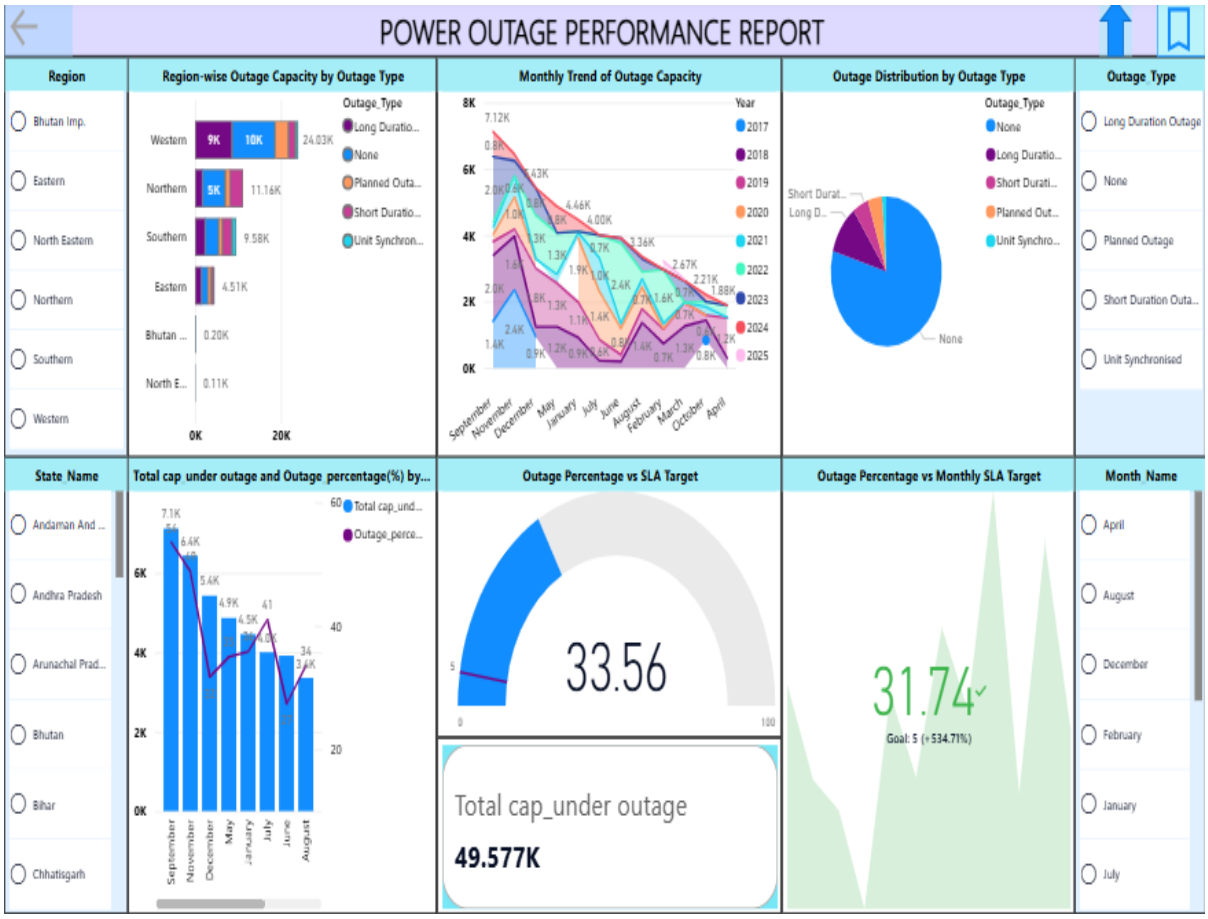
Fields Used:

- Category: Month_Name
- Breakdown: Outage_Type
- Values: Total_cap_under_outage

Insight: The waterfall chart clearly shows a progressive reduction in total outage capacity from September (~7.1K MW) to April (~2.7K MW), indicating a net improvement in outage control over time.



SCREENSHOT OF REPORT



9. Insights & Conclusions

Key Findings

- September recorded the highest total capacity under outage (~7.1K MW), making it the worst-performing month in terms of outage impact.
- October (~6.4K MW) and November (~5.4K MW) also showed significantly high outage capacity, indicating a continuous high-outage phase from September to November.
- December (~4.9K MW) and January (~4.5K MW) continued to show elevated outage levels, though slightly lower than the peak months.
- June, July, and August showed moderate outage levels (around 3K–4K MW), acting as a transition period before the peak outage months.
- February, March, and April recorded the lowest outage capacities, indicating relatively stable power supply during these months.
- The overall outage percentage is 33.56%, which is above the defined SLA target, mainly driven by high outages during September–January.

Analysis Insights:

Descriptive Analysis

- Month-wise analysis clearly shows that outage capacity is not evenly distributed across the year.
- The Monthly Trend of Outage Capacity visual confirms a sharp increase starting from August, peaking in September, and gradually reducing after January.
- The Line & Clustered Column chart highlights that months with higher outage capacity (September, October, November) also show higher outage percentages, directly impacting SLA performance.
- KPI and Gauge visuals reflect that the overall outage percentage (33.56%) exceeds the SLA threshold, influenced mainly by high-outage months.

Diagnostic Analysis

- The high outage capacity during September to November is largely driven by planned and long-duration outages, as seen in the outage-type distribution.
- Continuous high outages over multiple consecutive months (September–January) indicate scheduled maintenance concentration rather than isolated failures.
- Lower outage levels during February, March, and April suggest better system stability and reduced maintenance activities.

Predictive Analysis

- If current trends continue, September, October, and November are likely to remain high-risk months for SLA breaches in future years.
- The recurring pattern of increased outages from late Q3 to early Q4 suggests a seasonal maintenance cycle.
- Without rescheduling maintenance activities, outage percentage is expected to rise again during the same months.

Prescriptive Analysis

- Planned outages should be optimized or staggered during September–November to reduce cumulative outage impact.
- Critical maintenance activities should be shifted, where possible, to low-impact months such as February, March, and April.
- SLA monitoring should be strengthened during August itself, as an early warning before peak outage months.
- Region- and state-wise outage contributors during September–January should be prioritized for corrective actions.

10. Conclusions

The Power Outage Performance Dashboard clearly shows that the overall outage percentage of 33.56% exceeds the SLA target, primarily due to concentrated planned and long-duration outages between September and January, with September recording the highest outage capacity (~7.1K MW). The recurring spike from late Q3 to early Q4 confirms a seasonal maintenance-driven outage pattern rather than random failures. In contrast, February to April consistently show the lowest outage impact, proving that redistributing maintenance activities to these months can significantly reduce outage percentage and improve SLA compliance.